Geometry Review: Unit 7 — Area, Surface Area & Volume

Station 1 of 4: Area of Polygons and Composite Figures

Name: _____ Date: _____

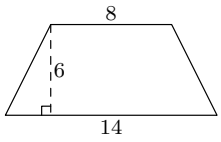
Include square units in your answers. Leave answers in terms of π unless told to round. Diagrams are not drawn to scale.

1. Find the area of a rectangle 9 cm by 14 cm.

2. Find the area of a triangle with base 10 in and height 7 in.

3. Find the area of a parallelogram with base 12 m and height 5 m.

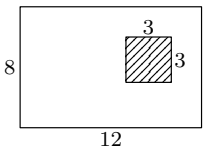
4. Find the area of a trapezoid with bases 8 and 14 and height 6.



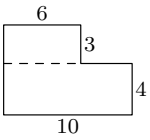
5. Find the area of a rhombus with diagonals 10 and 16.

6. A right triangle has legs 9 and 12. Find its area and perimeter.

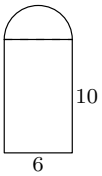
7. A 12×8 rectangle has a 3×3 square cut out of it. Find the area of the remaining figure.



8. An L-shaped figure is made of a 10×4 rectangle and a 6×3 rectangle that do not overlap. Find its total area.



9. A window is a rectangle 6 wide and 10 tall with a semicircle (diameter 6) on top. Find the area, to the nearest tenth.



10. A trapezoid has area 48 and bases 5 and 11. Find its height.

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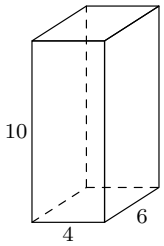
Station 2 of 4: Surface Area of Prisms and Cylinders

Name: _____ Date: _____

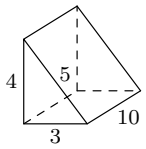
Include square units. Leave answers in terms of π unless told to round. Diagrams are not drawn to scale.

1. Find the surface area of a cube with edge 5.

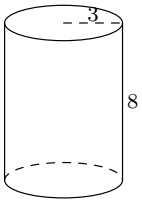
2. Find the surface area of a rectangular prism $4 \times 6 \times 10$.



3. A triangular prism has right-triangle bases with sides $3, 4, 5$ and length 10. Find its surface area.



4. Find the surface area of a cylinder with $r = 3$ and $h = 8$.



5. Find the lateral area of a cylinder with $r = 5$ and $h = 12$.

6. Find the total surface area of a cylinder with diameter 10 and height 7.

7. A cube has surface area 216. Find its edge length.

8. An open-top box (no lid) is $8 \times 5 \times 3$. Find the surface area of the five faces.

9. A can has radius 4 and height 10. Find the area of its label (lateral surface), to the nearest tenth.

10. A cylinder has radius 2 and surface area 24π. Find its height.

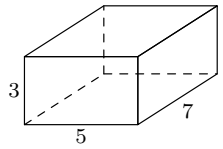
Geometry Review: Unit 7 — Area, Surface Area & Volume

Station 3 of 4: Volume of Prisms and Cylinders

Name: _____ Date: _____

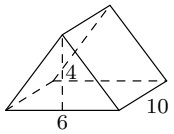
Include cubic units. Leave answers in terms of π unless told to round. Diagrams are not drawn to scale.

1. Find the volume of a rectangular prism $5 \times 7 \times 3$.

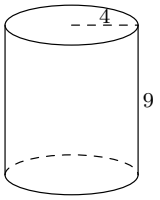


2. Find the volume of a cube with edge 6.

3. A triangular prism has a triangular base with base 6 and height 4, and the prism has length 10. Find its volume.



4. Find the volume of a cylinder with $r = 4$ and $h = 9$.



5. Find the volume of a cylinder with diameter 12 and height 5.

6. A rectangular prism has volume 90, length 5, and width 3. Find its height.

7. A cylinder has volume 75π and radius 5. Find its height.

8. A fish tank is 24 in long, 12 in wide, and 16 in tall. Find its volume.

9. A pipe is a cylinder with radius 2 cm and length 100 cm. Find its volume to the nearest tenth.

10. If the edge of a cube is doubled, by what factor does the volume increase?

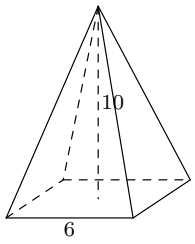
Geometry Review: Unit 7 — Area, Surface Area & Volume

Station 4 of 4: Pyramids, Cones, and Spheres

Name: _____ Date: _____

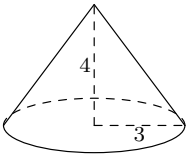
Use $V = \frac{1}{3}Bh$ for pyramids and cones. Leave answers in terms of π. Diagrams are not drawn to scale.

1. Find the volume of a square pyramid with base side 6 and height 10.

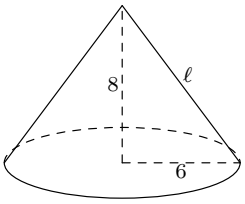


2. Find the volume of a rectangular pyramid with base 5×8 and height 9.

3. A cone has radius 3 and height 4. Find its volume and total surface area.



4. A cone has radius 6 and height 8. Find its slant height, lateral area, and volume.



5. Find the volume and surface area of a sphere with radius 3.

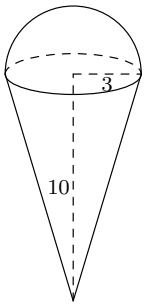
6. Find the volume and surface area of a sphere with radius 6.

7. A sphere has surface area 100π. Find its radius and volume.

8. Find the volume of a hemisphere with radius 4.

9. A cone has volume 64π and height 12. Find its radius.

10. An ice cream cone (radius 3, height 10) is topped with a hemisphere of radius 3. Find the total volume.



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Answer Key

Station 1: Area of Polygons and Composite Figures

1. 126 cm^2
2. 35 in^2
3. 60 m^2
4. 66 $\frac{1}{2}(8+14)(6)$
5. 80 $\frac{1}{2}d_1d_2$
6. 54; 36 *hypotenuse* $= 15$
7. 87
8. 58
9. $60+\frac{9}{2}\pi \approx 74.1$
10. 6 $48=\frac{1}{2}(16)h$

Station 2: Surface Area of Prisms and Cylinders

1. 150 $6\cdot 5^2$
2. 248 $2(24+40+60)$
3. 132 $2(6)+(3+4+5)(10)$
4. 66π $2\pi(3)(8)+2\pi(9)$
5. 120π
6. 120π $r=5$*:* $70\pi+50\pi$
7. 6
8. 118 $40+2(24)+2(15)$
9. $80\pi \approx 251.3$
10. 4 $2\pi(2)h+2\pi(4)=24\pi$

Station 3: Volume of Prisms and Cylinders

1. 105
2. 216
3. 120 $\frac{1}{2}(6)(4)\cdot 10$
4. 144π
5. 180π $r=6$
6. 6
7. 3
8. 4608 in^3
9. $400\pi \approx 1256.6\text{ cm}^3$
10. 8 2^3

Station 4: Pyramids, Cones, and Spheres

1. 120 $\frac{1}{3}(36)(10)$
2. 120
3. $V=12\pi;\ SA=24\pi$ *slant height* $=5$*;* $\pi(3)(3+5)$
4. 10; 60π; 96π
5. $V=36\pi;\ SA=36\pi$
6. $V=288\pi;\ SA=144\pi$
7. $r=5;\ V=\frac{500\pi}{3}$
8. $\frac{128\pi}{3}$
9. 4 $\frac{1}{3}\pi r^2(12)=64\pi$
10. 48π $30\pi+18\pi$